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Beyond deconstruction. a reconstructive perspective on sustainability transition governance

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ABSTRACT

This paper reviews criticisms of sustainability transition studies, using transition management (TM) as a case study. While these criticisms have yielded theoretical progress, underlying epistemological issues remain. Contrasting the TM approach to complexity with other more deconstructive views on complexity, it becomes clear that some criticisms on TM are inherently based on a deconstructive questioning of whether complex systems can be influenced into a desired direction. The authors build on those critiques to argue that TM needs to clarify how (1) TM itself harbours deconstructive power (hitherto insufficiently specified), while (2) at the same time having an explicit ambition to 'go beyond' deconstruction. To that end, this paper proposes a 'reconstructive approach' as an epistemological grounding for transition studies. This reconstructive approach is elaborated on three grounds: (1) a research focus beyond 'is' versus 'ought' towards 'can be', (2) interpretative research and reflexivity, and (3) a 'phronetic' understanding of sustainability.

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1. Introduction

The field of transition studies has taken up one of the great challenges in social science: understanding structural change. Transitions are defined as non-linear processes of social change, in which societal systems are structurally transformed (Markard et al., 2012; Grin et al., 2010; Geels and Schot, 2007). A 'sustainability transition' generally refers to a "radical transformation towards a sustainable society as a response to a number of persistent problems confronting contemporary modern societies" (Grin et al., 2010:1). One of the central premises in transition studies is that persistent problems are symptoms of unsustainable societies, and that dealing with these persistent problems in order to enable more sustainable systems, requires system innovations in specific sub-systems, as well as transitions that transcend individual systems and comprise various system innovations at different scale-levels and over long-term periods of time (Loorbach and Rotmans and 2010). A transition is the result of 'co-evolution'; "when the interaction between societal subsystems influences the dynamics of the individual subsystems, leading to irreversible patterns of change" (Grin et al., 2010: 4).

The oldest strand of transition research has its intellectual roots in innovation studies, as found in social studies of technology, where the multi-level perspective (MLP) was proposed as a framework to understand how 'regular', incremental innovation was structured by a 'regime', while niche experiments, regime changes and long–term contextual changes could produce a transition (Rip and Kemp, 1998; Schot, 1998; Geels, 2005). While originally the focus was on transitions in *socio-*

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technical systems fulfilling specific societal needs (e.g. systems for mobility, energy, agriculture), recent developments have extended the notion to other units of analysis, (e.g. regions, cities – Rotmans and Loorbach, 2010; Smith et al., 2010) and to 'reflexive' governance for sustainable development (Voß et al., 2009). In terms of analytical focus, the understanding of *transition processes* can be distinguished from the understanding of *how actors (can) influence* transition processes: the first object of study is referred to as *transition dynamics*, the latter as *transition management* (Rotmans et al., 2001; Loorbach, 2007; Loorbach and Rotmans, 2010) *transition governance* (Voß and Kemp, 2006; Grin, 2010; Frantzeskaki et al., 2012) or *strategic niche-management* (Hoogma et al., 2002). Yet, while they may be analytically distinguished, their development paths have been tightly intertwined.

Transition studies as a field is heavily under construction. Like many new fields of research, transition studies have encountered both enthusiasm and critical voices. This paper seeks to review some of this criticism with a keen eye to understanding its underlying assumptions, and to articulate implications for transition research. We start in section 2 by discussing various examples of criticisms on transition studies and its related sustainability governance discourse, using transition management as a case study. We conclude that, while some of the earlier criticism have been taken up and inspired theory development, underlying epistemological issues remain. A clearly articulated epistemological basis might help transition researchers to be more consistent and precise, especially in terms of how to conceive of the relationship between understanding transitions and influencing them; and in terms of the implications of that for *doing* transition management as a process of iterating between understanding and influencing. Such articulation may also help critics to better appreciate the ambitions and nature of transition management, stimulating more specific comments that may be productive for transition researchers in further developing the field. The latter is not to instrumentalise criticism, and even less to deny its intrinsic value; rather, we wish to emphasize that a young field needs to take up criticism, and that this may be promoted by accuracy on both sides.

Therefore in section 3, we move on to address epistemological assumptions of transition management literature and its underlying complexity paradigm. We contrast this with other views on complexity, in particular the view of deconstruction. We demonstrate how some criticisms on transition management are inherently based on a deconstructive logic. As we will further discuss in section 3, we understand 'deconstructive logic' here broadly as the critical scrutiny and unpacking of both societal and scientific practices and their underlying assumptions, such as the assumption that complex systems can be influenced into a desired direction. We then deliberate how to better benefit from this deconstructive critique on transition management. We will argue that interesting opportunities to build on this criticism emerge when we acknowledge that the criticized transition literature itself takes a deconstructive stance towards incumbent assumptions in social and scientific practices. More fundamentally, we will show that much work in transition studies crucially builds on – while also aiming to move beyond – a deconstructive position, towards what we call a *reconstructive* approach. Thus, we argue in section 4, transition studies could much more benefit from such criticism than its propagators have hitherto acknowledged, as it may help the effort to enable deconstruction of the social order as is, and then go beyond this towards the reconstruction of potential alternative futures. We elaborate this reconstructive approach on three grounds: (1) a research focus beyond 'is' versus 'ought' towards 'can be', (2) interpretative research and reflexivity, and (3) a phronetic understanding of sustainability.

2. Transition management: criticism & responses

2.1. Criticisms on early transition studies

For reasons of space, we focus our discussion on one key example of transition governance: the transition management (TM) approach. It is an interesting case for our purposes, as its development has been partly informed by practical experiences and applications (Loorbach and Rotmans, 2010). TM can be regarded as a governance model under development, which on its turn is being informed by theory development (Loorbach, 2007, 2010). It is continuously adapted and extended on the basis of explorative and design-oriented research on both *transition dynamics* and *transition governance*. Deductive research and theory development are combined with inductive and empirical research designs, including modelling approaches and action research. Complexity theory, governance theory and social theory are used to develop governance approaches, management 'instruments' and 'policy tools', which in turn are adapted on the basis of empirical testing and action research experiences.

One of the most common, and important, points of critique on early transition research has been the lack of attention to power and politics in transition literature and in TM literature in particular (Shove and Walker, 2007, 2008; Smith and Stirling, 2008; Hendriks, 2009; Meadowcraft, 2007; Meadowcroft, 2009; Stirling, 2011). As pointed out by Meadowcroft:

> "Transition management is not primarily concerned with the political processes through which societal goals are determined and revised, collective decisions are enforced, and resources are authoritatively allocated. Nor does it focus on the evolution of societal values and value conflict, and with spheres of individual and family life, the definition of group identities, and citizenship. Yet all these spheres are important to processes of societal change, and relevant to governance for sustainable development" (2007: 10).

Other authors have pointed out that "the tactical opportunities for the structural power of incumbent socio-technical regimes to mould discourse" challenge the "straightforward 'managerial' understandings of transition management and

sustainability governance", and that it is "unclear how these sit in relation to prevailing policy institutions and political processes" (Smith and Stirling, 2008: 12–13).

Some authors have also criticized that the notion of 'sustainability' is being used too unreflectively. Although the literature on TM emphasizes the ambiguous and subjective nature of sustainability, it is pointed out that still too little attention is given to the political implications of such ambiguity and the strategic games surrounding sustainability discourse (ibid, Smith and Kern, 2009). As Smith & Stirling (2008:15) frame this important question: "with notions of sustainability displaying such malleability to strategic interpretation, how credible is it that a transition management process that begins with a vanguard of elite visionary forerunners can really overturn structurally embedded regimes?"

The concern about strategic interpretation and 'malleability' does not only apply to sustainability discourse, but also to transition (management) discourse more generally. Shove and Walker state that "there is a politics to transition management, a playing out of power of when and how to decide and when and how to intervene, which cannot be hidden beneath the temporary illusion of 'post-political' common interest claims of sustainability" (2007:765-766). They emphasize that it is necessary to reflect on this politics of TM as "a now identifiable set of ideas around which actors and institutions explicitly orient themselves" and to ask 'what the label of transition management does', exactly (2008:1013). The authors even go as far as to state that the TM model *obscures* the "fluid and contested matters of boundary making and definitional power" (2008:1014).

Last but not least, authors express concerns about the involvement of researchers in transition politics. The TM model as it exists today has been 'co-produced' by researchers, policy-makers and other practitioners (Kemp and Rotmans, 2009; Loorbach, 2007) through action research in various policy fields. This has evoked criticism: "even though many researchers are undoubtedly perfectly capable of separating their double roles, it is well imaginable that this double role can obscure the analysis. The involvement with a policy practice or innovation project can possibly function as an impediment for a realistic analysis of the process. It makes it hard to make 'honest' claims about the role that the researcher himself has played in the process" (Duineveld et al., 2007: 26, *translated from Dutch*). This extends the above concerns about definitional power to researchers: not only may *societal actors* use transition discourse for 'strategic interpretation'; also *researchers* may use it to influence the framing of sustainability issues. Whether or not this is problematic in itself, critics emphasize that this role of researchers needs more critical scrutiny.

2.2. Reponses to criticism

Critical discussions on TM have prompted a manifold of empirically grounded studies on how to conceive of power, politics and discourse in transitions (e.g. Hendriks and Grin, 2006; Hendriks, 2009; Kern and Howlett, 2009; Voß et al., 2009; Grin, 2010; Avelino, 2009, 2011; Kern, 2011; Hoffman, 2013; Paredis, 2013; Pel et al., 2015). The criticism regarding the treatment of 'sustainability' have been addressed paying more attention to its nature as an essentially contested notion and response to persistent societal problems (Rotmans and Loorbach, 2010; Schuitmaker, 2012). More recently, methodical issues have been taken up by specifying TM as a transdisciplinary action research approach (Wittmayer and Schäpke, 2014).

While these studies have yielded highly needed enrichment and corrections to transition theory, many of the epistemological issues underlying the critique have as yet been under-addressed. Several of the criticisms, especially those on 'strategic interpretation', 'malleability', 'definitional power' and the dual role of transition researchers, rather explicitly raise concerns on the role of transition researchers (both in scholarly work and in engagement in transition governance). Being essentially questions "about the knower and what can be known" (Furlong and Marsh, 2010: 185), such critiques reveal epistemological viewpoints. In the remainder of this article, we seek to take up the challenge and discuss what this underlying critique may, and should, mean for transition studies.

First, we should point out that the transition research traditions as discussed in the introduction, when *understanding* transitions, go beyond a positivist ontology i.e. they go beyond realism and atomism. They do not reduce causation to direct, linear causal relationships between a force and an object; and they do not assume a deterministic kind of predictability of the system under study. As a consequence, such researchers, when discussing the *governance* of transitions, do *not* assume that the system may be simply 'controlled.' Rather, it seeks (i) to elaborate ways to attempts to 'influence', 'shape' or 'modulate' a system understood in a post-positivist, post-Newtonian way, while (ii) recognising that attempts to do so are essentially fallible and must hence be embedded in processes of learning and feedback.

A post-positivist approach is common to all traditions of transition research discussed in the introduction. In the socio-technical tradition, interactions within the regime, and between regime and niche experiments, are understood as multiple, sometimes indirect and not necessarily linear. They elaborate the notion of co-evolution while explicitly not reducing it to path dependency, including path creation and interlinking of multiple paths (e.g. Kemp et al., 2001; Geels and Schot 2007). The reflexive governance approach also takes an explicitly post-positivist perspective (see Voß et al., 2006, 2009), which has been elaborated more specifically in e.g. elaborations of power and empowerment (Avelino, 2009; Avelino, 2011; Hoffman, 2013; Grin, 2012) and dual track governance (Grin, 2010). In the next section, we will discuss these issues at more length for the example of transition management, which has explicitly been related to complexity theory (Loorbach, 2007: 51–100; Rotmans and Loorbach 2010: 140–160, Frantzeskaki et al., 2012).

3. Complexity, transition management, and deconstruction

3.1. The complexity paradigm as the basis for transition management

The complexity paradigm (De Haan and Rotmans, 2011), originating from mathematical studies of organic systems (Holland, 1995), has more recently entered social studies (e.g. Marion and Uhl-Bien, 2003; Pel, 2012; Room, 2011). Its different manifestations all address systems that exhibit behaviour that is not simply, or not at all, reducible to the interactions at the level of the system's composing parts (De Haan, 2010). We cannot explain phenomena like e.g. traffic jams by accumulating all the underlying interactions (i.e. between characteristics and behaviours of the road infrastructure, traffic regulation, cars, and drivers). A small and seemingly irrelevant small change in one element of a nonlinear system may result in large systemic changes, also referred to as the so-called 'butterfly effect' (Lorenz, 1969). This dynamic appeared to occur, for instance, after the 'traffic congestion scheme' of the London municipality, which, in unanticipated way, yielded novel patterning of social practices that thus reduced congestion (Shove and Walker, 2010). More specifically, complex *adaptive* systems (CAS) theory pays more explicit attention to systems' capacity to learn from experience and adapt themselves, responding to external changes. In TM literature, this is believed to "provide the necessary framework that allows us to include all characteristics of societal complexity, such as heterogeneous agents and artefacts, dualism and structures, emergence, surprise, and uncertainty" and to therefore be "very adequate to describe and analyse societal systems, while building on insight from sociology" (Loorbach, 2007:64).

Following the complexity paradigm, society may be understood as consisting of various 'complex adaptive systems' that form each other's environments. Internal dynamics of these systems are 'complex', and this complexity increases as these systems interact with their surroundings and with each other. Transition management has been developed as a combination of inductive understanding of empirical observations of and action research experiences with policy interventions, as well as deductive theoretical insights from sociology, governance research and complex systems theory. In 2001, Rotmans et al. characterised transitions as systemic shifts towards a new dynamic equilibrium, involving non-linear and (co-)evolutionary processes, and they presented transition management as an approach that is "more evolution than revolution" in that it "tries to utilize the opportunities for transformation that are present in an existing system" and "joins in with ongoing dynamics rather than forcing changes" (Rotmans et al., 2001:25). The underlying complex systems principles and the implications for governance were specified more elaborately in later work (e.g. Loorbach, 2007, 2010; Rotmans and Loorbach, 2010; Frantzeskaki et al., 2012).

Loorbach (2010) formulated governance principles based on complexity theory, including (amongst others): acknowledging how process and content are inseparable (process management requires content understanding of the system), using long-term thinking (over 25 years) as a framework for policy design, keeping policy objectives flexible and adjustable, playing into both equilibria and disequilibria and enabling multi-actor participation within the system. Frantzeskaki et al. (2012) refer to transition management in terms of a 'new form of incrementalism':

> "The premise of transition management is that (. . .) incremental changes in the societal system may turn into radical transformation in the long-term. This premise of transition management is theoretically deduced from the conceptualisation of the societal system as a complex adaptive system devoid of linearity between stimuli and responses; hence small-scale stimuli can have high impact, and bring about radical change. In an operational sense, the build-up of incremental changes in a stepwise matter with the vision of sustainability is an effort to interlink process and content in policy negotiations, it does not however imply control of the long-term process or assertiveness of the process outcome. The uncertainties and complexity inherent in long-term processes of societal change subvert such prescriptive approaches to change. The message and premise of transition management is that efforts to govern transitions to sustainability can be and need to be made despite the uncertainties and complexities in place"(Frantzeskaki et al., (2012)).

3.2. Different understandings of complexity

In order to clarify the epistemological positioning of transition management, and its relation to complexity, it is necessary to also acknowledge that there are different understandings of the notion of complexity, and that complexity as such can be related to different epistemological interpretations. For example, some authors have discussed links between the complexity paradigm, the notion of 'post-modernism' and deconstructivism (Cilliers 2005; Strand, 2002). Cilliers points out that Derrida (the 'founder' of deconstruction) has explicitly linked the fact that things are complex to the problems of meaning and context, and emphasized that "meaning and knowledge (. . .) are always contingent and contextual" and that "the context itself (. . .) has to be interpreted" (2005:25). Cilliers responds to some of the misconceptions about complexity, postmodernism, and deconstruction, and concludes that the acknowledgment of complexity does *not* need to result in vague or relativistic statements, but rather in a modest position towards knowledge; "we can make strong claims, but since these claims are limited, we have to be modest about them" (2005:263). Strand (2002: 164) has noted that with regard to governance, the complexity paradigm challenges the "belief in the strategy of reducing practical problems to a set of technical problems" and contends that we need to be critical and reflexive on how our understanding of complexity "affects and should affect our ideological basis for governance" (Strand, 2002:164).

So far, these views are in line with assumptions underlying TM, as described by e.g. Rotmans and Loorbach 2010 or Frantzeskaki et al., 2012. However, some deconstructive views on complexity go one crucial step further, contending that "the implication faithful to complex systems theory is *not* the question 'how can I govern the system into a new attractor (desired by me)?' but rather; 'what *is* my role in this system, and how *does* the action of me and others affect the system'?" (Stacey 2000 in Strand, 2002:176, *emphasis added*). Such views are typically critical of the belief underlying TM that we can or should 'steer' complex societal system towards a more desirable state. We argue that some of the critiques on TM are reminiscent of such deconstructive views. Shove and Walker, for instance, question the assumption that "better understanding will necessarily enhance the capacity to manage", and criticize the "very idea of transition management (. . .) that deliberate intervention in pursuit of specific goals, like those of sustainability, is possible and potentially effective" (2007:764).

These critiques respond to the claims of some TM scholars that "greater insight into the dynamics of a complex adaptive systems leads to improved insight into the feasibility of directing it", and that "the application of complexity theory can be used to direct complex, adaptive systems" and to transform "a complex adaptive system from one state to another" (Rotmans, 2005:34). Rotmans and Loorbach (2010:143) describe characteristics of complex systems and argue that "the challenge is to elaborate on these guidelines for managing complex systems while applying them to social systems (. . .) The question arises to what extent these simple but elegant principles can be applied to social systems. The underlying premise is that social systems are complex, adaptive systems. It further requires a one-to-one transposition between the formalized, deductive abstractions of complexity theory and the tenacious, complex real world".

These descriptions of the relations between TM and complexity, can be (mis)understood as a positivist translation from analysing dynamics to directing intervention. We argue that more epistemological nuance and clarification in the TM literature is necessary to emphasise that the notion of 'directing' here does not refer to traditional control from the outside, but that it refers to a type of control that is part and parcel of the complex system's dynamics. Although TM-scholars maintain that it is possible and desirable to analyse the dynamic mechanisms of complex societal systems and to use such analysis to promote a normatively favourable development, they do not at any point uphold that such knowledge is objective or will necessarily lead to improvement. As emphasized by Loorbach (2007: 65): "any focus on a societal system or subsystem is (. . .) arbitrary" and "there are no objective borders between sub-systems" (Loorbach, 2007:65). Any understanding being inherently uncertain, and complexity and non-linearity characterizing any intervention, an intervening actor can do no better than to base an intervention on such understanding, to monitor how the system evolves by way of adapting to an intervention, and to adapt her strategy accordingly. Thus, essentially, she becomes part of the complex adaptive system. This is the essence of self-governance, often confused with the idea that governance is not possible. Self-governance implies that the governance system is part of the governed system (Loorbach, 2007:165). As such, the complexity paradigm underlying transition management, inherently questions the possibility of there being a 'transition manager' in a unique position that can steer the system from outside.

3.3. Deconstruction and transition management

Now that we have elaborated the epistemological positioning of transition management and its relation to complexity, it is necessary to more carefully discuss other understandings of complexity, such as the deconstructive approach, and to subsequently discuss how this in turn relates to transition management. According to the founding father of deconstructionism, Derrida, a deconstructive approach "cannot exist" (Weitzner, 2007:43) and deconstruction cannot be defined: "all sentences of the type 'deconstruction is X' or 'deconstruction is not X" *a priori* miss the point, which is to say they are at the least false" (Gilbert Walsh 2007:318). Derrida does expand elaborately on what deconstruction *does*, namely it "interrupts a construction" (ibid:319). Such "interruption of constructions" has gained popularity in interpretative, post-positivist research traditions. Its task is to "negate the *illusion* of coherence, to expose how texts, when read carefully, deconstruct themselves and thereby reveal their lack of positive content" (ibid:321). Deconstruction, a term coined by Derrida ([1Derrida ((1967) [Derrida, 1978]1978) for a philosophical approach to reading text, later appeared to "take a life of its own (. . .) [as] it moved beyond the realm of 'philosophy proper' to permeate the discourse of the humanities and the social sciences, and eventually it settled into the common parlance of popular culture" (Gilbert-Walsh 2008:317). We use 'deconstructive logic' here broadly as a way to 'unpack' societal and scientific practices by articulating their underlying assumptions and showing with what other practices and (discursive, institutional or material) institutions these hang together. Thus seen, a deconstructive logic does, normatively, not necessarily reflect a post-modern perspective – it may as well be informed by, for instance, a critical-theoretical perspective.

In this basic meaning, even though most of the transition critics cited so far could not be sensibly classified as deconstructivists in the original meaning, some of the criticisms are inherently *based on a deconstructive logic.* Duineveld et al. (2007), for instance, have taken a figure from the TM literature that depicts "transitions as mutually enforcing societal radars", visualised as cog wheels (Rotmans, 2003:14). Having found this figure, the authors subsequently interpret the underlying implicit assumptions of such visualisation and argue that this depiction illustrates how TM is 'based on a technocratic, mechanistic view of society'. Other examples of critiques of TM that follow such deconstructive logic, include the following quotes:

> "The more we think about the politics and practicalities of reflexive transition management, the more complex the process appears: for a policy audience, our words of caution could be read as an invitation to abandon the whole endeavour. If agency, predictability, and legitimacy are as limited as we have suggested, this might be the only sensible conclusion." (Shove & Walker 2007:768)

> "A more important, more general function of process analysis is the production of 'negative suggestions' or deconstructions (. . .) There is no point in investing much time and money in research that produces [policy] instruments as results of research. 'Negative' suggestions can thus spare much energy, money and other capital, by mapping out the limits of manageability[1] beforehand" (Duineveld et al., 2007:33, *translated from Dutch*).

We argue that these type of critiques, partly based in a deconstructive logic, are a valuable form of scientific dialogue, drawing attention to understudied elements (such as power), challenging researchers to be clear and modest in their claims and reminding them to remain focused on a deliberative analysis. However, we also argue that when we take deconstruction seriously, it is necessary to also ask *why and what* we are deconstructing. Paraphrasing the question that Bernstein (1993) poses to post-modernist writings, we may ask: 'deconstruction in the name of what?' Bernstein (1993:6) argues that, following Wittgenstein's claim that all language games have a grammar, critique must have a standard – not in the positivist, foundational sense, but as a historically contingent point of departure that works through being interpreted in action contexts. An interesting text in this regard, concerns the following quote:

> "The critique of the inanities and injustices of present society, however obvious they may be, is disqualified by a simple reminder that remaking society by design may only make it worse than it is. Alternative ends are invalidated on the strength of the proved ineffectuality of means." (Bauman, 1991, quoted in: Shove and Walker 2007:763 in their article criticising transition management).

Powerful and relevant as this quote may be – warning us for unintended consequences and harmful paths paved with good intentions – there are ample hidden presuppositions in this text that in themselves beg to be deconstructed in order to prevent that undue implications are drawn from it. For instance, one might easily take this quote to imply that the only right decision is to not intervene at all. Is that what the authors are really promoting? We argue that deconstruction is an important and valuable part of scientific critique and interrogation. At the same time, it is necessary to ask 'deconstruction in the name of what?', and to remain critical of the new constructs that one is (intentionally or unintentionally) giving birth to in a deconstructive process – such as the above suggestion that alternative ends are – by default – invalidated by ineffectuality of means.

We also argue that TM itself harbours inherent deconstructive power, in terms of 'interrupting the construction' of regimes through deconstructing their underlying paradigms and their grip on societal systems. Here the object of deconstruction is directed at the practices and (discursive, institutional and material) structures constituting existing regimes. Hence, deconstruction is done 'in the name of' questioning existing regimes. This inherently includes a deconstruction of underlying scientific methods or approaches that legitimise such regimes (e.g. notions such as 'efficiency' or 'economic growth'). Often, 'regime deconstruction' goes far beyond a mere academic exercise and also results into instrumental suggestions that, at times, serve to reconstruct 'new regimes' that are believed to be 'better' by those who apply them. Yet, the underlying logic and method of deconstruction in itself, is similar. TM, and transition studies more generally, take critical deconstruction of existing regimes as a starting point. While continued critical interrogation of TM and other sustainability governance approaches is important as such, it is also useful to explore how transition studies can *help* analysts to deconstruct hidden assumptions, problems, and undesired effects of existing practices, systems and underlying scientific paradigms. As Duineveld et al. state:

> "Analyses of societal, political, and policy processes can be used to specify the hidden assumptions, problems and undesired effects of policy (. . ..) as ways to make the thoughts and actions of people and institutions into an object of continuous criticism. That (. . .) may well be the most important step for reflection and for change. Without criticism, policy gets tangled up in its own ways of thinking. Dryzek wrote that 'Resistance is fertile', thereby emphasizing that criticism and resistance are necessary to trigger (institutional) change. It is exactly that from which scientific research on (policy) processes can exert its non-scientific use and legitimacy." (Duineveld et al., 2007:34, *translated from Dutch*).

Thus, it is 'legitimate', if not quintessential, for researchers to deconstruct those very 'hidden assumptions, problems, and undesired effects of policy' so as to pre-empt policy being 'tangled up' by 'interrupting its construction'. We argue that TM itself, in not taking existing practices and structures for granted, essentially involves deconstruction of existing regimes, while at the same time, aiming to 'go beyond' deconstruction. In the next section, we will elaborate the implications of this to further elaborate some elements of an epistemology for transition governance research.

[1] Manageability is the chosen translation for the Dutch word 'maakbaarheid', literally 'make-ability'.

4. A reconstructive perspective on transition governance

While explicitly acknowledging the usefulness of deconstruction – in terms of deconstructing observed discourses on sustainability and transitions – we aim to elaborate on the TM ambition to go *beyond* deconstruction, by making explicit an approach that we call 'reconstruction'. By reconstruction we mean a process that (1) *starts with a phase of deconstructing,* i.e. 'interrupting' and questioning ingrained notions and practices, and is then (2) *followed by a phase of (re)constructing* new/adapted notions and practices, which (3) can then *replace/complement* those deconstructed notions and practices.

4.1. Beyond 'is' versus 'ought' towards 'can be'

To some readers, there might be a contradiction between this 'reconstructive' ambitions on the one hand, and the 'deconstructive' approach on the other hand. This relates to a fundamental debate in social science: between those who aim to 'describe' and 'explain', and those who wish to 'prescribe' and 'predict', between those who call for an understanding of how things 'are', and those who emphasize the understanding of how things 'ought to be'. Critical social theory in the trail of the Kant, Habermas and the Frankfurter Schule more generally, has predominantly occupied itself with the latter, openly defining its purpose as social research for social improvement. In the past decades, various social scientists have challenged this Enlightenment approach to critical theory and called for a re-appreciation of insights offered by Machiavelli, Nietzsche, and Foucault. Brocklesby and Cummings (1996) have proposed a shift from Habermas to Foucault as 'an alternative underpinning of critical social theory', emphasizing the advantage of understanding how things 'are' instead of focusing on how they 'should be'.

Similar pledges come from Flyvbjerg (1998), who emphasises the importance of understanding existing power relations and the intertwined nature of power and rationality. At the same time, however, Flyvbjerg (2001, 2004) has taken up the notion of phronèsis (Arendt, 1975) to argue for 'making social science matter again' by calling on researchers to dare make normative judgements about what 'ought' (not) to be, by answering the following 'phronetic' research questions: (1) where are we going; (2) who gains and who loses, and by which mechanisms of power; (3) is this development desirable; and (4) what, if anything, should we do about it? As such, Flyvbjerg combines 'is' and 'ought' approaches to social sciences. We argue that TM – and transition studies more generally – also combines 'is' and 'ought' approaches to research, while at the same time elaborating on these approaches by adding specific attention for research on what 'can be'.

The study of sustainability transitions is not only about studying how thing 'are' at a specific point in time, nor only about how they 'ought to be' in the future. Most crucially, it is (also) about understanding how things *can be*, now, in the near future, and in the long-term. In that light, the challenge is to analyse phenomena in terms of *transformative potential*, i.e. the capacity to change what 'is' in to that which is believed 'ought to be'. In its broadest sense, transition research by its nature is also about studying how things *can be*, and TM is about trying to operationalise a 'can-be-philosophy' in a political context dominated by 'is-versus-ought' debates.

As soon as we acknowledge that any 'manager' is part of the system, it is futile to aim to steer a system to a particular equilibrium state, chosen from a range of possible 'attractors' identified on basis of an analysis of the system 'as is'. Rather, novel attractors *are to be (re)constructed (!)* in multi-actor, deliberative practices, so as then, on their turn, provide orientation to further collective action. Similarly, the idea is not just to analyse the system so as to know its dynamics 'as is', but rather to promote dynamics that may help an envisioned attractor to 'become' through supporting and foregrounding 'frontrunners' (thus generating 'dissipative structures') as well as through creating and empowering 'niches' (stimulating emergence and co-evolution, respectively). Finally, given uncertainties about feasibility of identified 'attractors', and of the intended and unintended effects of promoting 'desired' system dynamics, variation and selection through experiments are included (Rotmans and Loorbach, 2010: 144–147)

Thus, the central challenge TM must meet in order to live up to these principles is to actually replace system analysis in terms of 'as is' by an exploration of the system 'as can be'. This critically hinges, of course, on *not* imposing analysis on the situation, but rather (i) using complex adaptive system notions to support stakeholders' deliberative practices in (ii) *contextually* analysing contemporary multi-level dynamics in relation to persistent problems so as to (iii) identifying *possible,* desirable attractors as well as actions that may help to realize them. At least as crucially is, to indeed consider TM as essentially part of a fallibilistic process of learning and experimenting, in which experiences are continuously fed back into the understanding of the system *as it could be*. Associate epistemological issues deserve much more attention in TM literature.

4.2. Interpretative research and reflexivity

This is where transition analysis and TM might benefit from better articulating its underlying epistemological position on a key question: how to avoid to 'choose sides' between 'is' versus 'ought' approaches to science? As Bernstein(1993:15-30) has argued, studying history may be a source of critique, as it may help to see how 'self-evident' practices, structures and underlying assumptions are historically contingent. Acknowledging this insight, well-established in critical-hermeneutic philosophy and social theory (e.g. Arendt, 1975; Emirbayer and Misch, 1998), allows for reflexive agency. Therefore, it is crucial – especially in transition studies – to explore the whole spectrum of what was, what is, what seems to be, what people want, and what we think that will be or ought to be. For that we need both deconstruction and reconstruction, both qualita-

tive and quantitative research, both positivistic and post-positivistic paradigms. As Fischer (1980, 1995) has argued, proper analysis brings together diverse epistemological perspectives, some strong at discussing causal relationships 'as are' (or, we add, as may be envisaged), others at contextualizing them to establish their meaning, still others scrutinizing underlying assumptions and the preferred nature of context. Somewhat paradoxically, this requires a particular post-positivist, interpretative and reflexive approach to scientific research in terms of ontological and epistemological starting points (Yanow and Schwartz-Shea, 2006; Fischer, 2007). Not to privilege that particular paradigm, but precisely because contrary to positivism (with its foundationalist assumption of but one truth, and its 'either-or' choice between realist foundationalism or relativism – Bernstein, 1988, 1993: 8 ff), the interpretative paradigm is at much more ease with different understandings of the world – and of our knowing of it.

More specifically, interpretative research aims at *reflexivity,* i.e. acknowledging the recursive relation between interpretation and the object of observation, and thus reflecting on one's own interpretation and role in (research) processes. Reflexivity "requires an awareness of the researcher's contribution to the construction of meanings throughout the research process, and an acknowledgment of the impossibility of remaining 'outside of' one's subject matter while conducting research" (Nightingale and Cromby, 1999:228). Researchers "cannot expect to hide behind a 'third-person' omniscient exposition – the so-called view from nowhere" (Schwartz-Shea, 2006:103). In transition studies, intervening actors are indeed not seen as exogenous to the system, but essentially as part of it. Moreover, reflexivity also gets an additional layer: researchers are interested in the potential for *transformation* of incumbent practices and their structural embedment. Thus reflexivity is not only treated as an *epistemological* issue but also as an *ontological* phenomenon, in the twofold sense that researchers need to be reflexive about their interpretations of and interventions in social reality, while at the same time acknowledging that actors and systems under study are also reflexive, thus harbouring and inherent potential for change (Grin, 2006).

Many authors have observed that TM can understand itself as a form of 'reflexive governance' that aims to explicitly integrate reflexivity (e.g. Kemp and Loorbach, 2006; Grin, 2010; Voß et al., 2006, 2007, 2009). Our corollary claim that this implies a place for deconstruction as a contribution to scrutinizing incumbent assumptions, however, has got less attention. Yet, this immediately raises the question of the limits of deconstruction – any reconstruction may, of course, always be deconstructed, but can we somehow tell how to balance between wise scrutiny and endless inactive hesitation?

4.3. Phronetic understanding of sustainability

Let us start at the extreme position. Some criticism on TM focuses on the critique that influencing a system from the perspective of sustainable development means to impose a specific normative objective. Such criticisms tends to disregard, however, that transition researchers approach sustainability as a concept that is intrinsically complex, normative, subjective, and ambiguous (Kasemir et al., 2003; Rotmans 2005) and context-specific (Grin, 2006). The notion of sustainability is a typical 'essentially contested notion', one of those concepts which "inevitably involve endless disputes about their proper uses on the part of their users", and "to engage in such disputes is itself to engage in politics" ([1Lukes (1974) [Lukes, 2002]2002:45). Sustainability is a potential answer to the question 'deconstruction in the name of what?' and – again referring to Bernstein (1993) – like any answer to that question, it is not universal but needs contextual interpretation to translate it into action. And different actors may – and will – interpret the notion in different ways.

In a specific action context, sustainable development may be more specifically defined in deliberation between the actors involved, and taking into the peculiarities of that context. Such pragmatic judgement, i.e. 'phronèsis', seeks to strike a contextually sensible balance between the reflective order (starting from understandings of what 'is' and preferences about the 'ought'), and the practical order (assumptions on what 'can be') (Loeber, 2007). This is never the end of reflection, but to be followed by further reflection-in-action. Reflected experiences gained may yield re-definition by actors who do not define themselves as external of the system, but as part of it and crucial to its adaptive and learning capacities (Grin, 2006). Deconstruction may be brought into the process, both from within the reflexive practice and from other spheres, and can inform such judgement. It may make the actors involved better aware of their own, and the others', assumptions (a basic condition for deliberation) and the ways in which these are contextually shaped and historically situated (a basic condition for reflexivity). Deconstruction may also trigger and inform later learning experiences.

A further warrant is that these processes of phronèsis are not proceeding in splendid isolation but essentially embedded. In contemporary, differentiated polities (Pollit, 2003), deliberative spheres like 'transition management arenas' or 'transition experiments' find themselves amidst a variety of other discursive spheres, from which ongoing processes of reconstruction may be continuously challenged. As Hendriks and Grin (2007) have shown, it is precisely at the interface between discursive spheres that legitimacy of (re-)constructions may be contested – and where *decisions* are eventually taken that may yield additional legitimacy to reconstructions. To be sure, decisions will always exclude some viewpoints, and that is precisely why continued deconstruction and reconstruction is a crucial trigger for and input in further processes of learning.

5. Conclusion

This paper has reviewed some of the criticism on early transition studies and its related sustainability governance concepts, taking transition management (TM) as a case-study. We argued that, while some of the earlier criticism have been taken up and inspired new studies and theory development, the underlying epistemological issues deserve more attention. We then demonstrated how some of the criticisms on TM and its underlying complexity paradigm are based on a decon-

structive understanding of complexity, which questions to what extent complex systems can be influenced into a desired direction.

In response to these deconstructive critiques, we have argued the following. First, TM itself harbours and presupposes inherent deconstructive power, which hitherto has been insufficiently specified, in terms of 'interrupting' the 'construction' of regimes by deconstructing regimes' underlying paradigms and their grip on societal systems. Second, TM at the same time also has an explicit ambition to 'go beyond' deconstruction. We elaborated the implications of these two points of departure to further formulate an epistemology for TM research. We have characterised this epistemology in terms of a 'reconstructive' approach, which (1) *starts with a phase of deconstructing,* i.e. 'interrupting' and questioning, ingrained notions and practices, and is then (2) *followed by a phase of (re)constructing* new/adapted notions and practices, which (3) can then *replace/complement* those deconstructed notions and practices. We elaborated this reconstructive approach with the following three arguments:

5.1. Beyond 'is' versus 'ought' towards 'can be'

Social science has known age-old debates between 'is-versus-ought' debates, between those that call for 'describing' and 'explaining' how things 'are' and those that emphasize 'prescribing' or 'predicting' how things 'ought to be'. TM – and transition studies more generally – combines 'is' and 'ought' approaches to research, while at the same time elaborating on these approaches by adding specific attention for research on what 'can be'. It is found crucial to explore the whole spectrum of what was, what is, what seems to be, what people want, and what we think that will be or ought to be. For that we need both deconstruction and reconstruction, both qualitative and quantitative research, both positivistic and post-positivistic paradigms.

5.2. Interpretative and reflexive research

Acknowledging the importance of different research paradigms, paradoxically requires a particular post-positivist, interpretative and reflexive approach to scientific research in terms of ontological and epistemological starting points. Contrary to positivism (with its foundationalist assumption of but one truth, and its 'either-or' choice between realist foundationalism or relativism), the interpretative paradigm it is at much more ease with different understandings of the world – and of our knowing of it. Such interpretation aims at reflexivity, i.e. to acknowledging the recursive relation between interpretation and the object of observation, and thus epistemologically reflecting on one's own interpretation and role in (research) processes, while at the same acknowledging ontologically that actors and practices under study are also interpretive and reflexive phenomena in themselves, and thus subject to change.

5.3. Phronetic understanding of sustainability

The field of sustainability transition research – including TM literature – acknowledges that the notion of sustainability is intrinsically complex, normative, subjective, ambiguous and context-specific. The notion of sustainability can be viewed as a potential answer to the question 'deconstruction in the name of what?', and like any answer to that question, it is not universal but needs contextual interpretation to translate it into action. In a specific action context, sustainable development may be more specifically defined in deliberation between the actors involved, and taking into the peculiarities of that context. Such pragmatic judgement, i.e. 'phronèsis', seeks to strike a sensible balance between the reflective order (understandings of what 'is' and preferences about the 'ought'), and the practical order (assumptions on what 'can be').

While TM emphasizes the need for stakeholder interaction, experimenting and monitoring, it could use more reflected guidelines on how to do so deliberatively and reflexively by *drawing on deconstructions*, both in terms of TM-researchers more explicitly engaging in deconstructive analysis as well as in terms of welcoming deconstructive critique by others.

Those who emphasize the sense of continued deconstructing criticism, on the other hand, may better distinguish between their own inclination to deconstruct as they like, and the all too easy objection that all reconstructions are equally wrong. We argue that this is 'all too easy' as it neglects that it does make a difference if reconstructions are rooted in processes of deliberative, transformative judgement, embedded in a process of trial and error learning within reach of other discursive spheres. Proposing, in generic terms, that science should merely criticize transition processes from the outside, denies or neglects this insight.

As such, a reflexive note to deconstruction is to articulate 'in the name of what' one deconstructs transitions and governance approaches. Some may do so in the name of promoting a more sustainable situations; others may discuss their analyses in the name of those who are or what is being excluded. Both contributions are essential in improving the human condition. Only where the 'in name of what?' question remains unanswered, deconstruction may become a mere disabler of action. As a corollary, even when we recognize that deconstruction must never stop, pragmatic judgement may help decide where it makes sense to nevertheless continue action, and take up continued deconstruction in the next round of iteration.

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